🚀 WebStarter CLI - Complete Installation Guide

This guide will walk you through setting up and running WebStarter CLI on your Windows PC.

📋 Prerequisites Checklist

Before starting, ensure you have:

- [ ] **Node.js** v14+ installed (Download)
- [ ] **PHP** 7.4+ installed (XAMPP or PHP)
- [ ] **Composer** installed (Download)
- [ ] **MySQL** or **MariaDB** (comes with XAMPP)
- [ ] **Code Editor** (VS Code, Sublime, etc.)

📦 Method 1: Install From Source (Recommended for Development)

Step 1: Download or Clone the CLI

```bash
# Option A: Download ZIP and extract to a folder
# Then navigate to that folder in terminal

# Option B: Clone from repository (if available)
git clone <your-repo-url>
cd webstarter-cli
```

Step 2: Install Dependencies

```bash
npm install
```

This installs:

- `inquirer` - Interactive CLI prompts
- `shelljs` - Shell command execution
- `adm-zip` - ZIP file extraction for icons

Step 3: Link Globally (Optional)

To use `webstarter` command from anywhere:

```bash
npm link
```

Now you can run:

```bash
webstarter
# or
create-webstarter
```

📦 Method 2: Install From NPM (Once Published)

If the package is published to NPM:

```bash
npm install -g webstarter-cli
```

Then run:

```bash
webstarter
```

🎯 Method 3: Run Without Installing

Use `npx` to run directly:

```bash
npx webstarter-cli
```

🔧 First-Time Setup

Step 1: Verify Node.js Installation

```bash
node --version
# Should show v14.0.0 or higher
```

Step 2: Verify PHP Installation

```bash
php --version
# Should show PHP 7.4 or higher
```

Step 3: Verify Composer Installation

```bash
composer --version
# Should show Composer version
```

If Composer is not found, download and install from getcomposer.org.

🚀 Creating Your First Project

Step 1: Run WebStarter CLI

```bash
webstarter
```

or if you're in the CLI directory without global install:

```bash
node index.js
```

Step 2: Follow the Prompts

```
╔═══ WebStarter CLI - PHP Generator ═══╗

✨ Your name (creator of projects): John Doe
📁 Project name: my-first-app
🔐 Include authentication? Yes
👨‍💼 Include admin panel? No
📧 Install PHPMailer? Yes
🎨 Download Phosphor Icons? Yes
```

Step 3: Wait for Generation

The CLI will:

1. Create folder structure
2. Generate PHP files
3. Create database schema
4. Download Phosphor Icons
5. Install PHPMailer via Composer

🗄️ Database Setup

Step 1: Start MySQL

If using XAMPP:

1. Open XAMPP Control Panel
2. Start **Apache** and **MySQL**

Step 2: Create Database

Navigate to your project folder:

```bash
cd my-first-app
```

Import the schema:

```bash
```

```
mysql -u root -p < dbschema/database.sql
```

Or use phpMyAdmin:

1. Open `http://localhost/phpmyadmin`
2. Click "Import"
3. Select `dbschema/database.sql`
4. Click "Go"

⚙️ Configure Environment

Step 1: Open `.env` File

```bash
code .env
# or open in any text editor
```

Step 2: Update Database Credentials

```env
DB_HOST=localhost
DB_NAME=my_first_app
DB_USER=root
DB_PASS=        # Leave empty if no password
```

Step 3: Configure Application URL

```env
APP_URL=http://localhost:8000
```

🌐 Start Development Server

Option 1: PHP Built-in Server

```bash
```

```
php -S localhost:8000
```

Then visit: `http://localhost:8000`

Option 2: XAMPP (Apache)

1. Move project to `C:\xampp\htdocs\my-first-app`
2. Visit: `http://localhost/my-first-app`

✅ Verify Installation

1. Test Homepage

Visit: `http://localhost:8000/pages/public/index.php`

You should see the welcome page.

2. Test Authentication (if enabled)

Visit: `http://localhost:8000/pages/public/login.php`

Try registering a new user.

3. Check Database

```sql
USE my_first_app;
SELECT * FROM users;
```

You should see the registered user.

🎨 Using Phosphor Icons

Icons are located in: `assets/icons/core-main/assets/`

Available Styles

- `regular/` - Regular weight icons
- `bold/` - Bold weight icons
- `light/` - Light weight icons
- `fill/` - Filled icons

- `duotone/` - Duotone icons
- `thin/` - Thin weight icons

Example Usage

```html
<img src="/assets/icons/core-main/assets/regular/user.svg" alt="User Icon">
<img src="/assets/icons/core-main/assets/bold/house.svg" alt="Home Icon">
```

📧 PHPMailer Configuration

Step 1: Get Gmail App Password

1. Go to Google Account Settings
2. Security → 2-Step Verification
3. App passwords → Generate new password
4. Copy the password

Step 2: Update `.env`

```env
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=your-email@gmail.com
SMTP_PASS=your-app-password
SMTP_FROM=noreply@yourapp.com
SMTP_FROM_NAME="My First App"
```

Step 3: Test Email

Create `test-email.php` in your project root:

```php
```

```php
<?php
require_once __DIR__ . '/vendor/autoload.php';
require_once __DIR__ . '/config/env.php';

use PHPMailer\PHPMailer\PHPMailer;

$mail = new PHPMailer(true);

try {
    $mail->isSMTP();
    $mail->Host = getenv('SMTP_HOST');
    $mail->SMTPAuth = true;
    $mail->Username = getenv('SMTP_USER');
    $mail->Password = getenv('SMTP_PASS');
    $mail->SMTPSecure = PHPMailer::ENCRYPTION_STARTTLS;
    $mail->Port = getenv('SMTP_PORT');

    $mail->setFrom(getenv('SMTP_FROM'), getenv('SMTP_FROM_NAME'));
    $mail->addAddress('test@example.com');

    $mail->isHTML(true);
    $mail->Subject = 'Test Email';
    $mail->Body = '<h1>It works!</h1>';

    $mail->send();
    echo 'Email sent successfully!';
} catch (Exception $e) {
    echo "Failed: {$mail->ErrorInfo}";
}
```

Run:

```bash
php test-email.php
```

🔧 Troubleshooting

Issue: "Composer not found"

Solution:

1. Download Composer from getcomposer.org
2. During setup, note the installation path (e.g., `C:\ProgramData\ComposerSetup\bin\composer.exe`)
3. When CLI asks for path, paste it
4. The CLI saves it for future use

Issue: "Database connection failed"

Solution:

1. Verify MySQL is running (XAMPP Control Panel)
2. Check `.env` credentials
3. Ensure database was created:

```bash
mysql -u root -p
SHOW DATABASES;
```

Issue: "Icons not downloading"

Solution:

1. Manual download: Visit phosphoricons.com
2. Download ZIP
3. Extract to `assets/icons/`

Issue: "Permission denied" on Windows

Solution: Run terminal as Administrator:

1. Right-click PowerShell/CMD
2. "Run as Administrator"
3. Navigate to project folder
4. Run commands

Issue: "Cannot find module 'inquirer'"

Solution:

```bash
```

```
npm install
```

📁 Project Directory Structure After Setup

```
my-first-app/
├── api/              ✅ API endpoints
├── assets/
│   ├── css/          ✅ Styles
│   ├── icons/        ✅ Phosphor Icons (if downloaded)
│   └── js/           ✅ JavaScript
├── config/           ✅ Configuration
├── dbschema/         ✅ SQL schema
├── pages/            ✅ Frontend pages
├── vendor/           ✅ Composer dependencies (if PHPMailer installed)
├── .env              ✅ Environment config
├── .htaccess         ✅ Clean URLs
├── index.php         ✅ Entry point
├── composer.json     ✅ Composer config (if PHPMailer)
└── README.md         ✅ Documentation
```

🎓 Quick Workflow Tutorial

1. Create Project

```bash
webstarter
```

2. Setup Database

```bash
cd my-project
mysql -u root -p < dbschema/database.sql
```

3. Configure

Edit `.env` with your settings

4. Start Server

```bash
php -S localhost:8000
```

5. Develop

- Add pages to `pages/`
- Add API endpoints to `api/`
- Add styles to `assets/css/`

6. Test

Visit `http://localhost:8000`

💡 Pro Tips

Tip 1: Use Multiple Projects

```bash
webstarter  # Creates project-1
cd ..
webstarter  # Creates project-2
```

Tip 2: Save Composer Path

The CLI remembers your Composer path after first use.

Tip 3: Version Control

```bash
git init
git add .
git commit -m "Initial commit from WebStarter CLI"
```

`.gitignore` is already configured!

Tip 4: Quick Database Reset

```bash
```

```
mysql -u root -p < dbschema/database.sql
```

This drops and recreates all tables.

📞 Need Help?

Common Commands

```bash
# Check Node version
node --version

# Check PHP version
php --version

# Check Composer
composer --version

# List npm global packages
npm list -g --depth=0

# Reinstall dependencies
npm install

# Update CLI
npm install -g webstarter-cli@latest
```

🎉 Success Checklist

After setup, you should have:

- [ ] WebStarter CLI installed
- [ ] Project created successfully
- [ ] Database imported
- [ ] `.env` configured
- [ ] Development server running
- [ ] Homepage accessible
- [ ] Login/Register working (if enabled)

- [ ] Icons available
- [ ] PHPMailer installed (if enabled)

🚀 **You're all set! Happy coding!**

For more information, check the main README.md file.